MTH-202 Discrete Mathematics Update MCQS For Quiz-2 File Solve By Vu Topper RM

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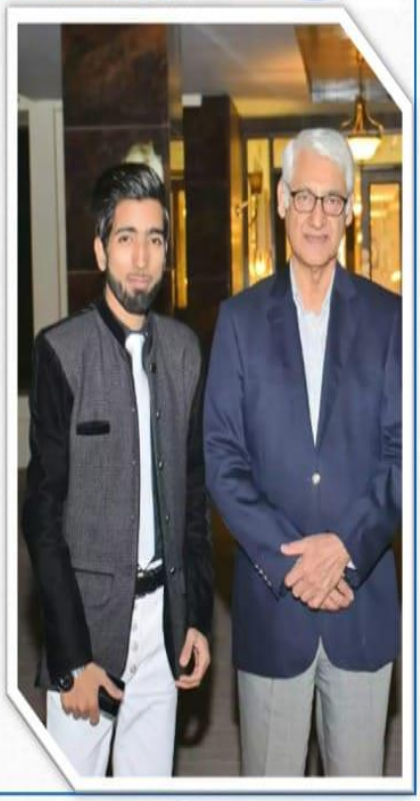
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Range of function \[f(x) = {e^x}\] is.....

A. Set of integers
B. Set of negative Real number.
C. Set of positive Real numbers
D. Set of Real number

The logic gate NOT is a uniary operation on {0,1}.

A. false
B. true

If f−1(x)=6−x2 then f−1(2) is .....

A. 2
B. 1
C. 0
D. 3

GIven a Set A={ a,b} and B={1,2,3}, then number of functions from A to B is .....

A. 6
B. 8
C. 5
D. 9

How many one-to-one functions are there from a set with five elements to a set with six elements.

A. 3 factorial (3!)
B. 4 factorial(4!)
C. 6 factorial(6!)
D. 5 factorial(5!)

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

A. true

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B. false

Which is not a binary operation on the set of natural numbers N?

A. Multiplication
B. Subtraction
C. None of these.
D. Addition

The relation R = {(12, 14), (13, 5), (-2, 7), (x, 13)} is a function if x is not equal to

A. 14, 7 or 13.
B. 12, 13 or 2.
C. 12, 13 or -2.
D. 14, 5 or 7.

Let $f(x)=x$ and $g(x)=-x$ for all $x \in R$, then $(f+g)(x)$ is____.

A. -x
B. 0
C. 2x
D. -2x

The Range of $f: X \rightarrow Y$ is also called the image of X under

A. true
B. false

Which of the following relations are functions? R1 ={(3, 4),(4, 5),(6, 7),(8, 9)}, R2 ={(3, 4),(4, 5),(6, 7),(3, 9)}, R3 ={(-3, 4),(4, -5),(0, 0),(8, 9)}, R4 ={(8, 11),(34, 5),(6, 17),(8, 19)}

A. R3 and R2 are functions
B. R1 and R3 are functions
C. R1 and R2 are functions
D. R2 and R4 are functions

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Cardinality of positive prime numbers less than 20 is …….

A. 5
B. 10
C. 8
D. 20

Which of the following is not a binary operation on the set of Integers?

A. Multiplication
B. Subtraction
C. Addition
D. Division

A set is countably infinite if,and only if,it has the same cardinality as the set of .....

A. whole numbers
B. real numbers
C. positive integers
D. integers

Let f be a function from X = {2, 4, 5} to Y = {1, 2, 4, 6} defined as f = {(2, 6), (4, 2), (5, 1)}. Then the range of f is _____ .

A. {2, 4, 5}
B. {1, 2, 6}
C. {1, 4, 6}
D. {1, 2, 4, 6}

A function f:R→R defined by f(x)=x−−√ is a real valued function.

A. true
B. false

A graph of a function f is onto iff every horizontal line intersects the graph in at least one point.

A. true

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B. false

Given a set X, define a function i from X to X by i(x) = x from all x belonging to X.Then _____ .

A. i is both injective and surjective
B. i is injective but not surjective
C. i is surjective but not injective
D. i is neither injective nor surjective

A constant function is one to one iff its …….. is a singelton.

A. Range
B. Domain

If f−1(x)=6−x2 then f−1(2) is .....

A. 2
B. 3
C. 1
D. 0

The logic gates OR and AND are uniary operation on {0,1}.

A. true
B. false

Let f(x)=x+3 then f−1(x) is .....

A. x+3
B. x−3
C. 3−x
D. 2x−3

The range of a function is always .....the co-domain of f.

A. a superset of
B. equal to
C. a subset of

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D. not equal to

The number of elements in A ×B are …… if A is a set with ‘5’ elements and B is a set with ‘4’ elements.

A. 0
B. 20
C. 1
D. 9

A function whose range consists of only one element is called…….

A. One to one function
B. Onto function
C. Constant function
D. Identity function

A set is countable if, and only if, it is _____.

A. infinite
B. finite or countably infinite
C. finite
D. countably infinite

If f(x)=2x and g(x)=x then g(f(x)) is.....

A. 2x
B. 2x2
C. x
D. x2

Let X = {1,2,3,4} and a function ‘f ‘defined from X to X by f(1) = 1 , f(2) = 1, f(3) = 1, f(4) = 1 then which of the following is true?

A. f is an identity function
B. f is injective
C. f is a constant function
D. f is bijective

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Domain of the relation {(0,1),(3,22),(90,34)}is _____ .

A. {1,22,34}
B. {0,1,3}
C. {0,1,3,22,90,34}
D. {0,3,90}

Let P(A) be the power set of a non-empty set A and by using the definition of "subset" relation, ⊆, for all X, Y ∈ P(A), X ⊆ Y ⇔ ∀ x, iff x ∈X then x ∈Y. Then _____.

A. ⊆ is a partial order relation.
B. ⊆ is not a partial order relation.

Let R be the universal relation on a set A then which one of the following statement about R is true?

A. R is not reflexive
B. R is not transitive
C. R is reflexive, symmetric and transitive.
D. R is not symmetric

Let A={1 , 2 , 3 , 4} and R={(1 , 1)(2 , 2)(3 , 3)(4 , 4)} then R is

A. Symmetric
B. Options a,b,and c all true
C. Reflexive
D. Transitive

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is.....

A. {(a,1)(2,b)(3,c)(4,d)}
B. {(1,a)(2,b)(3,c)(4,d)}
C. {(a,1)(b,2)(3,c)(4,d)}
D. {(1,a)(b,2)(3,c)(4,d)}

Range of the relation {(0,1),(3,22),(90,34)}is _____ .

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A. {0,3,90}
B. {0,1,3,22,90,34}
C. {1,22,34}
D. {0,1,3}

Let A = {1,2,3} and R be the relation defined on A such that R ={(1,1), (1,2), (2,3), (3,1)}, then R is _____ .

A. Symmetric
B. Antisymmetric

Let R and S be transitive relations on a set A then _____ .

A. Both R union S and R intersection S are transitive
B. R intersection S is transitive
C. Neither R union S is transitive nor R intersection S is transitive
D. R union S is transitive

There is atleast one loop in the graph of an irreflexive relation.

A. true
B. false

Let R be a binary relation on a set A.R is anti-symmetric iff _____.

A. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a = b
B. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a ≠ b
C. a, b ∉A if (a,b) ∈R and (b,a) ∈R then a ≠ b
D. None of the above

Find x and y given (2x, x + y) = (6, 2).

A. x=2,y=1
B. x=3,y=-1
C. x=2,y=-1
D. D. x=3,y=1 x=2,y=1

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Domain of a relation symbolically written as _____.

A. None of these
B. Dom (R) = {a ∈ A | (a, b) ∈ R}
C. Dom(R) = {b ∈ B | (a, b) ∈ R}
D. Dom (R) = {a ∉ A | (a, b) ∈ R}

Let A = {0,1}and B = {1}.Let R and S be two binary relations on Cartesian product of A and B such that R={(0,1)} and S ={(1,1)}.Then R intersection S = _____ .

A. {1,1}
B. {(0,1)}
C. {0,1}
D. empty

If x ≡ -10 (mod 15) Which of the following integers are valid solution for x ?

A. 7
B. 5
C. 3
D. 6

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Range of the inverse of the relation is _____.

A. {1, 3, 5, 7}
B. {5, 6, 7, 8}
C. {1, 2, 3, 4}
D. {2, 4, 6, 8}

Let R be a relation on a set A. If R is symmetric then its compliment is _____ .

A. Irreflexive
B. Symmetric

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C. Reflexive
D. Antisymmetric

If A={1 , 2 , 3} is a set and R = {(1 , 2) , (2 , 2) , (2 , 1)} is a relation on A, R is

A. Reflexive
B. Symmetric
C. None of these
D. Transitive

If a relation R={(1,1)(2,1)(1,2)(2,2)} is given then which of the following is not true about this relation.

A. R is Symmetric.
B. R is Transitive
C. R is Reflexive.
D. R is Irreflexive.

If x ≡ 17 (mod 5) Which of the following integers are valid solution for x ?

A. 4
B. -44
C. 8
D. 12

Let R be a relation on a set A. If R is reflexive then its compliment is _____ .

A. Reflexive
B. Antisymmetric
C. Symmetric
D. Irreflexive

Let the set A = {1,2,3,4} Then the relation {(2,4),(4,2)} is _____.

A. Transitive

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B. AntiSymmetric
C. Reflexive
D. Symmetric

The union of the transitive binary relations R = {(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)} and S = {(2,1), (2,4), (3,3)} on the set A = {1,2,3,4} is given by R ∪ S = {(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)}then R ∪ S is_____?

A. not transitive
B. transitive

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Domain of the inverse of the relation is _____.

A. {2, 4, 6, 8}
B. {1, 2, 3, 4}
C. {1, 3, 5, 7}
D. {5, 6, 7, 8}

Let A = {1,2,3,4} and define the relation R on A by R = {(1,2), (2,3), (3,3), (3,4)}.Then _____

A. R is neither reflexive nor irreflexive
B. R is both reflexive and irreflexive
C. R is reflexive
D. R is irreflexive

Range of a relation symbolically written as _____.

A. None of the above
B. Ran (R) = {a ∈ A | (a, b) ∈ R}
C. Ran (R) = {b ∈ B | (a, b) ∈ R}
D. Ran (R) = {b ∉ B | (a, b) ∈ R}

The matrix representation of an inverse relation can be obtained by taking the .....of the relational matrix.

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A. transpose
B. Determinant
C. inverse
D. adjoint

Let R be a relation on a set A. R is transitive if and only if for all a,b,c∈A then:

A. (a, b) ∈R and (b, c) ∈R but (a, c) ∉R
B. (a, b)∈R then (b, a) ∉R
C. (a, b) ∈R and (b, c) ∈R then (a, c) ∈R
D. (a, b) ∈R and (b, c) ∈R then (a, c) ∈R(a, b)∈R then (b, a) ∈R

The properties of being symmetric and being anti-symmetric are _____.

A. Negative of each other
B. Not negative of each other

Let A = {1, 2, 3, 4} and define a relation R on A by R = {(1, 1), (1, 2), (1, 3), (2, 3)}.Then which one of the following is a correct statement about R:

A. R is both transitive and reflexive
B. R is neither reflexive nor transitive
C. R is transitive
D. R is reflexive

Let A = {0, 1, 2} and R = {(0,2), (1,1), (2,0)} be a relation on A. Then which of the following statement about R is true?

A. R is reflexive
B. R is symmetric
C. None of the above
D. R is transitive

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Which of the following is always true for the matrix representation of a symmetric relation?

A. Matrix has always 0 in its diagonal entries
B. Matrix has always 1 in its diagonal entries
C. Matrix is singular
D. Matrix is equal to its transpose.

If A=(1,2,3) & B=(4,5,6) and R={(1,4)(2,5)(3,6)(3,4)} The complementary relation is .....

A. A × B
B. A × B (intersection) R
C. A × B (difference or -) R
D. A × B (Union) R

Let A = {1,2,3,4} and R = {(1,2), (2,3), (3,3), (3,4)} be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

A. (3,3)
B. (3,4)
C. (2,3)
D. (1,2)

Range of function /[f(x)=(e^x)/] is _____

- **Set of positive real numbers**

Composite relation symbolically written as _____

- **SoR={(a,c)aeA, ceC, 3eB, (a,b)eR and (b,c)eS}**

If x=17(mod 5) which of the following integers are valid solution for x ?

- **12**

Range of the relation {(0,1),(3,22),(90,34)}

- **{1,22,34}**

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Let A= {0,1,2} and R = {(0,2),(1,1),(2,0)} be a relation on A. The which of the following ordered pairs are needed to make it transitive?

- (0,0) and (2,2)

Operation of subtraction is a binary operation on the set of _____

- Integers

Let S=R and define the 'square' relation R= $\{(x,y) | x^2=y^2\}$. The square relation is an _____ relation

- Equivalence relation

The logic gate NOT is a uniary operation on {0,1}.

- True

Let A= {1,2}, then P(A)=_____

- {{},{1},{2},{1,2}}

If a relation R is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

- Inverse relation will be irreflexive.

Let R be a binary relation on a set A,R is anti-symmetric iff _____

- a,beA if (a,b)eR and (b,a)eR then a=b

"-" is a binary operation on the set of integers Z.

- True

The inverse relation R^(-1) from B to A is defined as _____

- R^(-1) = {(b,a) e B*A I (a,b)eR}

Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix is equal to its transpose

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Let A= {1,2,3,4} and let R and S be transitive binary relations on A defined as; R= {(1,2), (1,3), (2,2), (3,3), (4,2), (4,3) and S={(2,1), (2,4), (3,3)} then RuS= {(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)}

- R union S is transitive

Let S=R and define the ' square' relation R = {(x,y)Ix^2=y^2}. The square is an _____ relation.

- Equivalance relation

If x=-10(mod 15). Which of the following integers are valid solution for x?

- 5

Let R be a binary relation on a set A. If R is anti symmetric then _____

- Inverse of R is anti symmetric

If A={1,2,3} is a set and R = {(1,2),(2,2),(2,1)} is a relation on A, R is

- Symmetric

Let A={0,1} and B=(1). Let R and S be two binary relations on Cartesian product of A and B such that R={(0,1)} and S= {(1,1)}. Then R intersection S=_____

- Empty

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- Partial order Relation

Let X={1,2,3} and Y={7,8,9} and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- Co-domain of f must contain 1 element

The function fog and gof are always equal

- False

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If a relation R={(1,2),(2,3),(3,4)(4,1)(2,2) is given then which of the following is true about this relation.

- R is reflexive

A set is called countable if , and only if, it is_____

- finite

Let $f(x) = x^2-1$ define function f from R to R and c=2 be any scalar, then c,f(x) is _____

- $2x^2+2$

The set Z of all integers is _____

- Countable

Let R be a binary relation on a set A. If R is anti symmetric then _____

- Inverse of R is symmetric

For (2x-3, 4y+2) = (1,10). What will be the value of x and y ?

- (2,2)

Let f and g be the two functions from R to R defined by f(x) = IxI and g(x)= square root of x^2 for all xeR. Then_____

- F(x) is not equal to g(x)

If a set A has 15 elements then P(A) (power set of A) has _____ elements.

- 2^15

For the relation below to be a function, x cannot be what values {(12,14),(13,5),(-2,7),(x,13)}?

- X cannot be 12, 13, or -2

Let the set A = {1,2,3,4}. Then the relation {(2,4),(4,2)} is _____

- Symmetric

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For the following relation to be a function, x can not be what values? R={(2,4),(x,1), (4,2),(5,6)}.

- x cannot be 2,4 and 5

Vertical line test is used to determine that whether the graph of a relation is a function or not.

- True

The properties of being symmetric and being anti symmetric are _____

- Not negative of each other

The number of elements in AxB are _____ if A is a set with '5' elements and B is a set with '4' elements.

- 20

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is _____

- {(1,a)(2,b)(3,c)(4,d)}

Logic gate NOT does not define a binary operation on (0,1) because _____

- It takes a single input and gives a single output

How many real numbers exist between 1 and 5

- 3

The number pi is

- Irrational

The number square root 2 is

- Irrational

Range the relation {(0,1)(3,22),(90,34)}

- {0,3,90}

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Supherical coordinate 0 is related to the cylindrical coordinate as_____

- @=@

Operation of subtraction is a binary operation on the set_____

- Integers

Let A {1,2,3,4} and R={(1,2),(2,3),(3,3),(3,4)} be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

- (3,3)

Input values of the function are called the _____

- Domain

Range of function f(x)=IxI will be

- Set of positive real numbers

Which of the following is not a binary operation on the set of integers?

- Division

In the matrix representation of an irreflexive relation all the entries in the main diagonal are _____

- 0

If the partition set of A is $\{A_1,A_2\}$ then

- $A_1 n A_2$= not empty set

Let A= {a,b} then P(A) =

- {Non empty set, {a},{b},{a,b}}

Which relations below are not functions?

- {(13,14),(13,5),(16,7),(18,13)}

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In the directed graph of an antisymmetric relation there is _____ pair of arrows between two distinct elements of the set.

- **No**

If a relation R= (1,1),(2,1)(2,2) is given then which of the following is not true about this relation

- **R is irreflexive**

Let R and S be transitive relations on a set A then _____

- **Neither R union S is transitive nor R intersection S is transitive**

Let R={(1,2)(3,4)(5,6)(7,8)}. Domain of the inverse of the relation is _____

- **{2,4,6,8}**

Let A={1,2,3,4,5} and B={4,9,,16,17,25}. Then the relation R={(2,4),(3,9),(4,16),93,17)} The inverse of R is'

- **{(4,2),(9,3),(16,4),(17,3)}**

Let R be a relation on a set A. If R is symmetric then its compliment is _____

- Irreflexive

Which is not a binary operation on the set of natural numbers N?

- **Subtraction**

If a relation R={(1,1)(2,1)(1,2)(2,2)} is given then which of the following is not true about this relation.

- **R is irreflexive**

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is _____

- **{(1,a)(2,b)(3,c)(4,d)}**

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For any set A, the Cartesian product of A and A is known as _____

- Universal relation

Let A={p,q,r,s} and define a relation R on A by R={(p,p),(p,r),(q,r),(q,s),(r,s)} Then which one of the following is the correct statement about R:

- R is not reflexive

A={1,2} B={3,4}, R={(1,3)(2,4)}. Then the complement of R is _____.

- {(1,4)(2,3)}

Domain of a relation symbolically written as_____.

- Dom(R)={aeAI(a,b)eR}

Let X={2,4,5} and Y={1,2,4} and R be a relation from X to Y defined by R={(2,4)(4,1)(a,2)}. For what value of 'a' the relation R is a function?

- 5

Let A={1,2,3,4,5,6,7,8,9}, then which of the following sets represent the partition of the set A?

- A={1,3,5,7,9}, B={2,4,6}, C={8}

Let A={1,2,3} and B={2,4} then number of binary relations from A to B are _____.

- 64

A relation R is said to be _____iff it is reflexive, antisymmetric and transitivde.

- Partial order Relation

Let f be a function from X={2,4,5} to Y={1,2,4,6} defined as:f={(2,6),(4,2),(5,1)} . The range of f is _____

- {1,2,6}

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Let A={0,1,2} and R={(0,2),(1,1),(2,0)} be a relation on A. Then which of the following statement about R is true?

- R is symmetric

Let A={2,3,4} and B={2,6,8} and let R be the "divides" relation from A to B i.e for all (a,b) belong to (Cartesian product of A and B), a,R b iff a I b (a divides b). Then

- R={(2,2),(2,6),(2,8),(3,6),(4,8)}

Let A={1,2,3,...,50} and B={2,4,6,8,10}. Then the Cartesian product of A and B has _____ elements.

- 250

In the matrix representation of an reflexive relation all the entries in the main diagonal are _____

- 1

Which of the following is not a type of a relation?

- Permutation

Let X={2,4,5} and Y={1,2,4} and R be a relation from X to Y defined by R={(2,4),(4,1),(a,2)}. For what value of 'a' the relation R is a function ?

- 5

Which of the following is not a representation of a relation?

- Venn diagram

Let A={1,2,3,4} and define the following relations on A. Then R={(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)} is _____

- R is irreflexive

The range of f:X-> Y is also called the image of

- True

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Complementary Relation symbolically written as _____

- R=A*B- R={(a,b)eA*BI (a,b)not belong to R}

Let A={1,2,3,4} and R=(1,1)(2,2),(3,3),(4,4) then R is

- All options

If a relation R={(1,1),(2,1),(1,2),(2,2)} is given then which of the following is not true about this relation.

- R is irreflexive

Which of the following logical connective is not a binary operation ?

- Implication

A set may be dividend up into its disjoint subsets, such division is called_____

- Partition

If A=(1,2,3)&B=(4,5,6) and R ={(1,4)(2,5)(3,6)(3,4)} The complementary relation is _____

- A*B(difference or –) R

In matrix representation of a_____ relation, the diagonal entries are always 1.

- Reflexive

R is not symmetric iff there are elements a and b in A such that _____

- (a,b) belongs to R but (b,a) does not belong to R

Which relations below are functions?

R1={(3,4),(4,5),(6,7),(8,9)}

R2={(3,4),(4,5),(6,7),(3,9)}

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R3={(-3,4),(4,-5),(0,0),(8,9)}

R4={(8,11),(34,5),(6,17),(8,19)}

- R1 and R3 are functions

The logic gate OR and AND are uniary operation on {0,1}

- False

There is atleast one loop in the graph of an irreflexive relation

- False

There is atleast one loop in the graph of an reflexive relation

- True

A contains 3 elements and B contains 2 elements, then number of subsets of A*B are _____

- 64

Let A={1,2,3} and B={0,1,2} and C={a,b} R={(1,0),(1,2),(3,1),(3,2)} S={(0,b),(1,a),(2,b)} composite of R and S=_____

- {(1,b),(1,a),(3,a),(3,b)}

If R is transitive then the inverse relation will be transitive

- True

The number of elements in the power set of P(not empty set) denoted by P(P(not empty set) is

- 2

The function defined from Z to Z as f(x)= 1/(x+2)(x-2) is not well defined because _____

- Function is not defined at x=2 and x=-2

Rang of relation {(0,1),(3,22),(90,34)} is _____

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- {1, 22, 34}

The number of elements in the power set of P(not empty set) denoted by P(not empty set) is

- 1

Let R={(1,2),(3,4),(5,6),(7,8)}. Domain of inverse of the relation is _____

- {2,4,6,8}

The relation 'divides' on the set of integers is _____

- A symmetric relation

Operation of subtraction is a binary operation on the set of_____

- Integers

If R is transitive then the inverse relation will be transitive.

- True

Let A={1,2,3,4} and define the relation R on A by R={(1,2),(2,3),(3,3),(3,4)}. Then_____

- R is both reflexive and irreflexive

A set may be divided up into its disjoints subsets,such division is called

- Partition

If a set A contains is elements then the number of elements in its power set P(A) is _____

- 2^n

Range of a relation symbolically written as _____.

- Ran R = {b ∈ B I (a, b) ∈ R }

Let R be a binary relation on a set A. IF R is anti symmetric then _____.

- Inverse of R is anti symmetric

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Let A be a set with m elements and B be a set with n elements then the number of elements in A*B are _____

- m.n

Let A{1,2,3,4} and define the following relation on A. Then R={(1,3),(2,2),(2,4),(3,1),(4,2)} Is _____

- R is symmetric

If A={1,2,3}& B={4,5,6} and R={1,4},{2,5},{3,6},{3,4}. The complementary relation is _____

- A*B (difference or -) R

Let R be a relation on a set A. If R is reflexive then its compliment is _____

- Irreflexive

25=1(mod 3) means that 3 divides _____

- 25-1

Let R be the universal relation on a set A then which one of the following statement about R is true ?

- R is reflexive, symmetric and transitive

Domain of a relation symbolically written as _____

- Dom(R)= {aeR I (a,b) e R}

Let R be a relation on a set A. R is transitive if and only if for all a,b,ceA then

- (a,b)eR and (b,c)eR then (a,c)eR

Let A be a non-empty set and P(A) the power set of A Deifne the ' subset' relation ,c as follows for all X,Y e P(A), XcY <-> for all x, iff xeX then xeY . Then c is _____

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- c is partial order relation

Define a relation R={(1,1),(2,2),(3,3),(1,3) the relation is

- R is reflexive and transitive

Let R be a relation on a set A. R is transitive if and only if for all a,b,ceA then.

- (a,b)eR and (b,c)eR then (a,c)eR

Let R and S be reflexive relations on a set A then R intersection S is reflexive

- True

A function whose range consists of only one element is called_____

- One to one function

The set Z of all integers is _____

- Countable

One-to-one correspondence means the condition of _____

- Both (a) and (c)

Let X={1,5,9} and Y={3,4,7}, Define a function f from X to Y such that f(1)=7,f(5)=3,f(9)=_____. Which is true f(9) to make it a one-to-one (injective) function?

- 4

What will be the fourth term of the following sequence 1/2,2/3,3/4 _____?

- 4/5

The value of 6!=

- 720

A constant function is one to one iff its _____ is a singleton.

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- **Domain**

A constant function is onto iff its _____ is a singleton.

- **Co-domain**

Number of one to one functions from X={a,b} to Y={u,v} are equal to _____-

- **2**

If f is defined recursively by f(0) = -1 and f(n+1)=f(n)+3, then f(2)=_____.

- **5**

A function whose inverse function exists is called a/an_____

- **Invertible function**

Let f(2)=3, g(2)=3, f(4)=1 and g(4)=2 then the value of fog(4) is.....

- **3**

Let A={1,2,3,4} and B={7} then the constant function from A to B is _____.

- **Onto**

Composition of a function is a commutative operation.

- **False**

Composition of a function is not a commutative operation

- **True**

The sum of first five whole number is _____.

- **10**

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- **True**

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Inverse of a surjective function is always a function.

- **False**

Inverse of a surjective function may not be a function

- **True**

Let X={1,2,3,4} and Y={7,8,9} and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- **Co-domain of f must contain 3 elements**

If f; X->Y and g; Y->Z are both onto functions. Then gof; X->Z is _____

- **onto**

If f and g are two one-to-one functions, then their composition gof is_____

- **One-to-one**

If f: W →X, g:X →Y, and h:Y →Z are functions, then_____

- **(hog)of = ho(gof)**

Cardinality of positive prime numbers less than 20 is _____.

- **8**

IF $f(x)=\sin^{-1}(x)$ and g(x)=sin x then gof(x) is _____.

- **X**

0!=_____.

- **1**

An important data type in computer programming consists of _____.

- **Finite sequences**

Let f(x)=2x and g(x)=x+2 define functions f and g from R to R, then (f-g)(x) is _____.

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- x-2

The total number of terms in an arithmetic series 0+5+10+15+....50 are _____.

- 11

9!/6!=_____

- 504

Let f(x)=3x and g(x)=3x-2 define functions f and g from R to R, THEN (F+G)(X) is _____

- 6x-2

If f is a bijective function then ($f^{-1}f(x)$) is equal to

- X

A sequence whose terms alternate in sign is called an_____

- Alternating sequence

Common ration in the sequence "4, 16, 64, 256,..." is.....12.

- 4

0.8181818181 is a infinite geometric series.

- True

The word 'algorithm' refers to a step-by-step method for performing some action.

- True

A predicate become _____ when its variables are given specific values

- Sentence

The sum of two irrational numbers must be an irrational number.

- False

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The sum of two irrational numbers need not be irrational number

- **True**

The division by zero is allowed in mathematics.

- **Fasle**

The product of any two consecutive positive integers is divisible by 2

- **True**

If ‘n’ is an odd integer then n^3+n is _____.

- **Even**

For integers a,b,c, If divides b and a divides c, then a divides (a+b).

- **False**

Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that _____ and 0<r<d

- **N=d.q+r**

A rule that assigns a numerical value to each outcome in a simple space is called

- **Random variable**

If A and B are two disjoint (mutually exclusive) events then P(AB)=

- **P(A) + P(B)**

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

- **C(10,5)**

The expectation of x is equal to

- **Sum xf(x)**

If P(A intersection B) = P(A) P(B) THEN THE events A and B are called

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- Independent

A walk that starts and ends at the same vertex is called.

- None optins

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

- 497

What is the probability of getting a number greater than 4 when a die is thrown?

- 3/5

Eater formula for graphs is_____.

- F=e-v+2

X+a,x+3a,x+5a.....is an_____

- Arithmetic sequence

Composition of a function is a commutative operation.

- False

Real valued function is a function that assigns ____ to each member of its domain.

- Only a real number

Let X={1,2,3,4} and a function 'f' defined on X f(1)=1,f(2)=2,f(3)=3,f(4)=4 then _____

- F is an identity function

A constant function is surjective if and only if _____

- The co-domain consists of a single element

Cardinality means the total number of elements in a set.

- True

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If f(x)=2x and g(x)=x then g(f(x)) is _____.

- $2x^2$

Let f:R->R is one to one function then c,f, c is not equal 0 is also one to one function.

- True

Let X={1,2,3,4} and a function 'f' defined from X to X by f(1)=1, f(2)=1, f(3)=1, f(4)=1 then which of the following is true?

- F is a constant function

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- True

Which of the following is not correct for a 'sequence'?

- A sequence is a relation whose domain is the set of natural numbers

$F(x)=x^2$ is not one to one function from R to R^+

- True

Let f: R->R is one to one function then c,f c is not equal to is also one to one function.

- True

Let $f(x)=x+2$ then $f^{-1}(x)$ is_____

- x-2

Let $f(x)=x^2+1$ define functions f from R to R and c=2 be any scalar, then c,f(x) is _____.

- $2x^2-1$

One to one correspondence means the condition of _____.

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- Both (a)and (c)

A function F: R-> R defined by f(x) = square root x is a real valued function.

- False

If g:R->R defined by g(x)=e^2 is a real valued function of a real variable.

- True

A function F:R-> R defined by f(x) = log x is a real valued function

- True

½, then 3rd term of sequence is _____.

- 1/2

The process of defining an object in terms of smaller versions of itself is called recursion.

- True

Which of the following is not correct for a 'sequence'?

- A sequence is a relation whose domain is the set of natural numbers

A set is called countable if, and only if, it is _____.

- Finite and countable infinite.....both

A set that is not countable is called _____.

- Uncountable

A sequence whose terms alternate in sign is called an _____.

- Alternating sequence

Let f: R->R is one to one function then c,f is also one to one function for _____.

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- C is not equal 0

Let f(x) = x+3 then $f^{-1}(x)$ is _____-

- x-3

Let f and g be two functions defined by f(x) = x+2 and g(x)= 2x+1. Then the composition of f and g is_____.

- 2x+3

Number of one to one functions form X={a,b} to Y={u,v} are equal to _____

- 2

The flbonacci sequence is deined as $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$ for all integers $k \geq 2$ then which of the following is true for F_2

- $F_2-F_1= 2+1=3$

x+a, x+3a, x+5a,..... is a/an_____.

- Arithmetic sequence

Inverse of a function may not be a function.

- True

In the following sequence $a_k=K/(k+1)$, for k=1, a_1 will be _____.

- ½

If f(x)=x and g(x)= -x are both one to one function then (f+g)(x) is also one to one function.

- False

If f(x)=x and g(x)= -x are both one to one function then (f+g)(x) is not one to one function.

- True

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The function ‘f’ and ‘g’ are inverse of each other if and only if their composition gives_____.

- **Identity function**

Which of the following set is the domain of a sequence?

- **Set of real numbers**

Let C is defined as the set of all countries in the world then C is a _____.

- **Finite set**

A constant function is surjective if and only if_____.

- **The co domain consists of a single element**

The sum of the series $a_1 + a_2 + a_3 + \ldots\ldots$ can be written as _____.

-

Inverse of a function may not be a function.

- **True**
- **3**

Let X={1,2,3,4} and a function ‘f’ defined from X to X by f(1)=1, f(2)=1, f(3)=1, f(4)=1 then which of the following is true?

- **F is a constant function**

The composition of function is always

- **Associative**

A set is countably infinite if, and only if, it has the same cardinality as the set of

- **Positive integers**

Two functions ‘f’ and ‘g’ from ‘X’ to ‘Y’ are said to be equal if and only if _____.

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- F(x)=g(x) for all ‘x’ belongs to X

Y=x^3 is a graph of bijective function from R to R.

- True

Domain and range are same for_____.

- Identity funtion

Let X={1,2,3,4} and a function ‘f’ defined on X by f(1)=1,f(2)=2,f(3)=3

- F is an identity function

Composition of a function is a commutative operation.

- True

Inverse of a surjective function is always a function.

- False

A function whose inverse function exists is called a/an_____.

- Invertible

Given a set X define a function I from X to X by i(x)=x from all x belonging to X . Then_____.

- I is both injective and surjective

Let f: R->R is a one to one function then c,f is also one to one function for.

- C is not equal 0

Let X={1,5,9} and Y={3,4,7}. Define a function f from X to Y such that f(1)=7, f(5)=3, f(9)=_____. Which is true for f(9) to make it a one-to-one (injective) function?

- 4

Which of the following is not a predecessors of ak?

- A_k+1

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Two functions ‘f’ and ‘g’ from X to Y are said to be equal if and only if_____.

- **F(x)=g(x) for all ‘x’ belongs to x**

The two functions ‘f’ and ‘g’ are equal if _____.

- **F(x) =3x and g(x)= $6x^2+3x/2x^2+1$ for all xeR**

If first term of a geometric sequence is 2 and common ratio is ½, then 3rd ter, of sequence.

- **¼**

Y= squre root x is an _____function form R^+ to R

- **One to one function**

Y= x^2 is an _____function form R to R^+

- **NOT ONE TO ONE FUNTION**

If a function (gof)(x) : X->Z is defined as (gof)(x)=g(f(x)) for all xeX, Then the function_____—.

- **(gof)**

If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is _____.

- **-20**

If $f(x)=\sin^{-1}(x)$ and g(x) = sin x then gof(x) is _____.

- **X**

F: X->Y that is both one to one and onto is called a _____.

- **Bijective function**

What does ‘y’ denotes in a geometric sequence?

- **Common ratio**

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Let g be a function defined by g(x)=x+1. Then the composition of (gog).

- X+2

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Which of the following is true for the following sequence?

- If n is even, then Cn=2 and if n is odd, then Cn = 0

The function 'f' and 'g' are inverse of each other if and only if their composition gives_____.

- Identity function

N! is defined to be _____.

- The product of the integers from 1 to n

Let A = {1,2,3,4} and B={7} then the constant function from A to B

- Both one to one and onto

A set is called countable if, and only if, it is _____.

- Countably infinite and finite

If f(x) = x and g(x) = -x are both one to one functions then (f+g)(x) is also one to one function.

- False

If a is the 1st term and d be the common difference of an arithmetic sequence then the sequence is a, a+d, a+2d, a+3d....

- True

x+a, x+3a, x+5..... is a/an_____.

- Arithmetic sequence

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inverse of an injective function may not be a function.

- **True**

$y=x^3$ is a graph of bijective function form R to R

- **False**

Two functions 'f' and 'g' from x to y are said to be equal if and only if _____.

- **F(x)=g(x) for all 'x' belongs to x**

Common ration in sequence '36, 12, 4 , 4/3, ...' is .....

- **1/3**

A set is called finite if, and only if, is the _____ or there is ———.

- **Empty set or one-to-one**

Let $f(X)=x^2-1$ and g(x)=x+1 define functions f and g from R to R, then (f/g)x

- **x-1**

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- **True**

Let g be a function defined by g(x) = x+1. Then the composition of (gog).

- **X+2**

F:X->Y that is both one to one and onto is called a _____.

- **Bijective function**

What does 'l' denotes in a geometric sequence?

- **Common ratio**

Let A={1,2,3,4} and B={7} then the constant function from A to B is _____.

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- **Both one to one and onto**

N! is defined to be _____.

- **The product of the integers from 1 to n**

A set is called countable if, and only if, it is _____.

- **Both b and c**

Which of the following is the example of an alternating sequence?

- **C_n=n/n+1 for n $\geq$ 0**

X+a, x+3a, x+5a..... is an_____

- **Arithmetic sequence**

0, -5, -10, -15, ... is an _____.

- **Arithemetic sequence**

5, 9, 13, 17, ... is an _____.

- **Arithemetic sequence**

An important data type in computer programming consists of _____.

- **Finite sequence**

One-to-one correspondence means the condition of _____.

- **One-one and onto ....both**

Cardinality means the total number of elements in a set.

- **True**

Inverse of a function may not be a function.

- **True**

The functions 'f' and 'g' are inverse of each other if and only if their composition gives _____.

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- **Identity function**

A set is called finite if , and only if, it is the _____ or there is _____.

- **Empty set or one-to-one**

Let f and g be the two functions from R to R defined by f(x) = IXI and g(x) = square root x^2 for all xeR, then _____.

- **F(x) is not equal to g(x)**

If $f^{-1}(x) = 6-x/2$ then $f^{-1}(2)$ is _____.

- **2**

Let f(2)=3, g(2)=3, f(4)=1 and g(4)=2 then the value of fog(4) is _____.

- **3**

An important date type in computer programming consists of _——.

- **Finite sequence**

Let f(x)=x+3 then $f^{-1}(x)$ is _____.

- **X-3**

If f:X->Y amd g:Y->Z are both onto function. Then gof : X->Z is _____.

- **One-to-one function**

The functions 'f' and 'g' are inverse of each other if and only if their composition gives

- **Identity function**

The two function 'f' and 'g' are equal if _____.

- **F(x) = 3x and g(x) = $6x^2$+3x/2x+1 for all xeR**

Two functions 'f' and 'g' from X To Y are said to be equal if and only if——.

- **F(x) and g(x) for all 'x' belongs to X**

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Which of the following set is the domain of a sequence?

- **Set of natural numbers**

If 1st term of a geometric sequence is 2 and common ratio is 1/2 , then 3rd term of sequence is _____.

- **1/2**

Composition of a function is a commutative operation.

- **True**

If a function (gof)(x): X-> Z is defined as (gof)(x)=g(f(x)) for all xeX. Then the function _____ is known as composition of f and g.

- **(g o f)**

A set is countably infinite if and only if and only if, it has the same cardinality as the set of _____.

- **Positive integers**

The 3rd term of the sequence $b_n=5^n$ is _____.

- **125**

If f is a bijective function then $(f^{-1}(f(x))$ Is equal to ____.

- **X**

Let f(x) = x and g(x) = -x for all xeR, then (f+g)(x) is ____.

0

An infinite sequence may have only a finite number of values.

- **True**

The functions fog and gof are always equal .

- **False**

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If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- True

A function whose range consists of only one element is called _____.

- Constant function

Let X ={1,5,9} and Y={3,4,7}. Define a function f from X to Y such that f(1)=7, f(5)=3, f(9)=4 then which of the following statement about 'f' is true?

- F is both one-to-one and onto

$Y=x^3$ is a graph of bijective function from R to R.

- True

A function whose inverse function exists is called a/an_____.

- Invertible

Let F and g be two functions defined by f(x)= x+2 and g(x)= 2x+1. Then the composition of f and g is ____.

- 2x + 5

If r is a positive real number, then the value of r in 3, r,r=-27r is

- -9

The ____ of the terms of a sequence forms a series.

- Sum

The sum of first five whole number is _____.

- 10

If $f_k=f_k-1+f_k-2$then $f_0=1$, $f_1=2$, then f_2= _____.

- 3

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Let f(x) = 3x and g(x) = x + 2 define functions f and g from R to R, then (f.g)(x) is _____.

- $3x^2 + 6x$

Let R be a relation on a set A. If R is reflexive then its compliment is _____ .

- Irreflexive

If A = Set of students of virtual university then A has been written in the _____.

- **Descriptive form**

If a function (g o f)(x):X→Z is defined as (g o f)(x) = g(f(x)) for all x ∈ X. Then the function _____ is known as composition of f and g.

- **(g o f)**

If X and Y are independent random variables and a and b are constants, then Var(aX + bY)is equal to

- **aVar(X) + bVar(Y)**

Let A = {1, 2, 3} and B = {2, 4} then number of functions from A to B are _____.

- **8**

p is equivalent to q' means _____.

- **p is necessary and sufficient for q.**

Let A and B be subsets of U with n(A) = 12, n(B) = 15, n(A') = 17, and n(A intersection B) = 8, then n(U)=_____ .

- **29**

For the following relation to be a function, x can not be what values? R = {(2,4), (x,1), (4,2), (5,6)}

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- **x cannot be 2, 4 or 5**

Find the number of the word that can be formed of the letters of the word "ELEVEN".

- **120**

There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

- 6

Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

- **16**

A predicate becomes _____ when its variables are given specific values.

- **statement**

Find the number of distinct permutations that can be formed using the letters of the word "BENZENE"

- **420**

Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- **40**

In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

- **24**

If p ↔ q is True, then _____.

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- **p and q both are True.**

If A and B be events with P(A) = 1/3, P(B) = 1/4 and P(A ∩ B) = 1/6, then P(A ∪ B) = _____ .

- **5/12**

an integer n is a perfect square if and only if _____ for some integer k.

- **n = k^2**

If A and B are disjoint finite sets then n(A ∪ B) = _____.

- n(A) + n(B)

Let X = {2, 4, 5} and Y= {1, 2, 4} and R be a relation from X to Y defined by R = {(2,4), (4,1), (a,2)}. For what value of 'a' the relation R is a function ?

- **5**

~(P → q) is logically equivalent to _____.

- **p ∧ ~q**

A tree is normally constructed from _____.

- **left to right**

A Random variable is also called a _____.

- **Chance Variable**

The conjunction p ∧ q is True when _____.

- **p is True, q is True**

The logical statement p ∧ q means _____.

- **p AND q**

Which of the followings is the factorial form of 5 . 4?

- **5!/3!**

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What is the minimum number of students in a class to be sure that two of them are born in the same month?

- **13**

If p is false and q is true, then ~p ↔ q is _____.

- **True**

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- **TRUE**

(p ∨ ~p) is the _____.

- **Tautology**

(-2)! = _____ ?

- **Undefined**

If p = It is raining, q = She will go to college
"It is raining and she will not go to college"
will be denoted by

- **p ∧ ~q**

Let X = {1, 2, 3}, then 2-combinations of the 3 elements of the set X are _____?

- **{1, 2}, {1, 3} and {2, 3}**

Let $f(x) = x^2 + 1$ define functions f from R to R and c = 2 be any scalar, then c.f(x) is _____.

- **$2x^2 + 2$**

The disjunction of p and q is written as _____.

- **p ∨ q**

If X and Y are independent random variables, then E(XY) is equal to

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- E(x)E(y)

How many possible outcomes are there when a fair coin is tossed four times?

- 16

Which of the followings is the product set A * B * C ? where A = {a}, B = {b}, and C = {c, d}.

- {(a, b, c), (a, b, d)}

The number of the words that can be formed from the letters of the word,"COMMITTEE" are

- 9! / (2!2!2!)

One-to-One correspondence means the condition of _____.

- One-One and onto

The functions f o g and g o f are always equal.

- FALSE

If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- K-Sample

Determine values of x and y, where (2x, x + y) = (8, 6).

- x = 4 and y = 2

Let g be a function defined by g(x) = x + 1. Then the composition of (g o g)(x)is _____.

- x + 2

What is the truth value of the sentence?
'It rains if and only if there are clouds.'

- False

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Reductio and absurdum' is another name of _____.

- Proof by contradiction

X belongs to A or x belongs to B, therefore x belongs to _____.

- A union B

Which of the followings is the product set A * B * C? Where A = {a}, B = {b}, and C = {c, d}.

- {(a, b, c), (a, b, d)}

Real valued function is a function that assigns _____ to each member of its domain.

- Only a real number

The negation of "Today is Friday" is

- Today is not Friday

A non-zero integer d divides an integer n if and only if there exists an integer k such that _____.

- n = d k

The statement $p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$

- True

Let R be the universal relation on a set A then which one of the following statement about R is true?

- R is reflexive, symmetric and transitive.

Let f(x)=3x and g(x) = 3x − 2 define functions f and g from R to R. Then (f+g)(x) = _____.

- 6x − 2

The switches in parallel act just like _____.

- OR gate

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The converse of the conditional statement p → q is

- **q → p**

If X and Y are random variables, then E(aX) is equal to

- **aE(X)**

Which of the following statements is true according to the Division Algorithm?

- **17 = 5 x 3 + 2**

Let p → q be a conditional statement, then the statement q → p is called _____.

- **Converse**

The disjunction p ∨ q is False when _____.

- **P is False, q is False.**

A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

- **30**

The converse of the conditional statement 'If I live in Quetta, then I live in Pakistan' is _____.

- **If I live in Pakistan, then I live in Quetta.**

The functions 'f' and 'g' are inverse of each other if and only if their composition gives _____.

- **Identity function**

P (0, 0)=_____?

- **1**

Let p1, p2, p3 be True premises in a given Truth Table. If the conjunctions

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of the Conclusion with each of p1, p2, p3 are True, then the argument is _____.

- Valid

If p is false and q is false, then ~p implies q is _____.

- False

A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement?

- 125

Let A = {2, 3, 5, 7}, B = {2, 3, 5, 7, 2}, C = Set of first five prime numbers. Then from the following which statement is true ?

- A = B

The set of prime numbers is _____.

- Infinite set

The contrapositive of the conditional statement 'If it is Sunday, then I go for shopping' is _____.

- I do Not go for shopping, then it is Not Sunday.

Let p be True and q be True, then (~p ∧ q) is _____.

- False

In how many ways a student can choose a course from 2 science courses,3 literature courses and 5 art courses.

- 30

The method of loop invariants is used to prove _____ of a loop with respect to certain pre and post-conditions.

- correctness

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A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions

- **126**

If A and B are any two sets, then A − B = B – A

- **False**

There are 5 girls students and 20 boys students in a class. How many students are there in total?

- **25**

A sequence whose terms alternate in sign is called
Series

Cauchy sequence

Alternating sequence

None of the above

The 3rd term of the sequenceB5=5n

125

625

5

25

If a function is recursively defined as f(0) = 4 & f(n + 1) = 3f (n) + 3 then f(2)=.....

15

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55

48

50

The _____ of the terms of a sequence forms a series.

Sum

Multiplication

Let f:R->R is one to one function then c,f, c is not equal 0 is also one to one function.

False

True

720 = ...

4!

5!

7!

6!

The composition of functions is always _____.

Onto

Associative

one-to-one

commutative

What will be the fourth term of the following sequence 1/2 , 2/3 , 3/4 , ___?

5/6

7/8

4/5

6/7

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Which of the following set is the domain of a sequence?

Set of natural numbers

Set of complex numbers

Set of real numbers

Set of rational numbers

Let f(X)=x2-1 and g(x)=x+1 define functions f and g from R to R, then (f/g)x

X2-1

X2+1

x-1

x+1

9!/6!=_____.

506

509

504

509

If a function is recursively defined as f(0) = 4& F(n + 1) = 3f (n) + 3 then f(1) =____

25

10

20

15

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